

Explanation: Every cell in SDRAM and DDR are refreshed every 64 milliseconds. But trying to refresh everything together will result in a power surge which is not good, so it is staggered. 128 MB and smaller modules are refreshed every 15.6 microseconds. 256 MB modules every 7.8 microseconds (μsec). Today's RAM modules can handle more than the recommended 64 μsec between refreshes. A setting of 128 μsec increases performance and reduces power consumption.

SDRAM Command Leadoff

Options: 3, 4

Recommended Setting: Depends on system stability

Explanation: 3 is better.

SDRAM Idle Limit/SDRAM Idle Timer

Options: Disabled, 0 cycles, 8 cycles, 12 cycles, 16 cycles, 24 cycles, 32 cycles, 48 cycles

Recommended Setting: 12 cycles for less than 512 MB, 32 cycles for all others

Explanation: The idle time before SDRAM recharge. Higher values postpone the recharge and reduce RAM latency. Use this together with the Refresh Interval setting for best results.

SDRAM Precharge Control/SDRAM Page Control

Options: Enabled, Disabled

Recommended Setting: Enabled. Disable it if there is instability.

Explanation: This checks whether the RAM or the chipset should control the refreshing. Enable for RAM to control the refresh. Can cause instability with large amounts of RAM or poor-quality RAM.

Video RAM Cacheable

Options: Enabled, Disabled

Recommended Setting: Disabled

Explanation: Enabling this causes video data to be cached in the L2 cache. Can cause a performance bottleneck when enabled.